



JABchem



Not to be shared without the copyright holder's permission

Past Papers Nat 5 Chemistry

2015 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/100)	%	
A	68+	68%	32.3%
B	58+	58%	20.8%
C	48+	48%	19.4%
D	43+	43%	8.5%
No award	<43	<43%	19.0%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	13.4 /20	31.1 /60	14.1 /20

2015 National 5 Chemistry Marking Scheme											
MC Qu	Answer	% Pupils Correct	Reasoning								
1	A	92	Atomic number = number of protons $\therefore$ atomic number = 26 Mass number = number of protons + number of neutrons $\therefore$ mass number = 26 + 30 = 56								
2	B	88	Isotope Definition: same number of protons atomic number different number of neutrons mass number ☒ A W and X have different numbers of protons $\therefore$ W and X not isotopes ☒ B W and Y have same number of protons but different numbers of neutrons ☒ C X and Y have different numbers of protons $\therefore$ X and Y not isotopes ☒ D Y and Z have different numbers of protons $\therefore$ Y and Z not isotopes								
3	D	67	☒ A Cl^- ions have an electron arrangement of 2,8,8 ☒ B S^{2-} ions have an electron arrangement of 2,8,8 ☒ C Ar atoms have an electron arrangement of 2,8,8 ☒ D Na^+ ions have an electron arrangement of 2,8								
4	C	55	☒ A Gas would not need to pass through water to escape ☒ B Gas would not pass through water (test tube would explode as pressure builds) ☒ C Gas passes through water and escapes test tube through tube at top ☒ D Gas would pass through water but test tube would explode as pressure builds								
5	D	78	☒ A this structure shows metallic bonding ☒ B this structure shows ionic bonding ☒ C this structure shows molecular covalent bonding ☒ D this structure shows covalent network bonding								
6	C	51	<table border="1"> <tr> <td>Chromium chloride has the formula</td><td>Chloride ions have the formula</td><td>3 chloride ions per chromium chloride</td><td>Chromium ion must have 3+ charge to balance charge</td></tr> <tr> <td>CrCl_3</td><td>Cl^-</td><td>$\text{Cr}^{n+}(\text{Cl}^-)_3$</td><td>$\text{Cr}^{3+}(\text{Cl}^-)_3$</td></tr> </table>	Chromium chloride has the formula	Chloride ions have the formula	3 chloride ions per chromium chloride	Chromium ion must have 3+ charge to balance charge	CrCl_3	Cl^-	$\text{Cr}^{n+}(\text{Cl}^-)_3$	$\text{Cr}^{3+}(\text{Cl}^-)_3$
Chromium chloride has the formula	Chloride ions have the formula	3 chloride ions per chromium chloride	Chromium ion must have 3+ charge to balance charge								
CrCl_3	Cl^-	$\text{Cr}^{n+}(\text{Cl}^-)_3$	$\text{Cr}^{3+}(\text{Cl}^-)_3$								
7	C	46	☒ A Calcium at surface as it is less dense (1.54g cm^{-3}) than calcium chloride (2.15g cm^{-3}) ☒ B Calcium would be a solid at 800°C as it melts at 842°C ☒ C Solid calcium would float on the surface of the molten calcium chloride ☒ D Calcium would be a solid at 800°C as it melts at 842°C								
8	B	90	$2\text{Al}_{(\text{s})} + 3\text{Br}_{2(\text{l})} \longrightarrow 2\text{AlBr}_{3(\text{s})}$								
9	A	69	☒ A $\text{gfm SO}_2 = (1 \times 32) + (2 \times 16) = 32 + 32 = 64\text{g} \therefore \text{mass} = n \times \text{gfm} = 0.2 \times 64 = 12.8\text{g}$ ☒ B $1\text{mol CO} = (1 \times 12) + (1 \times 16) = 12 + 16 = 28\text{g} \therefore \text{mass} = n \times \text{gfm} = 0.2 \times 28 = 5.6\text{g}$ ☒ C $1\text{mol CO}_2 = (1 \times 12) + (2 \times 16) = 12 + 32 = 44\text{g} \therefore \text{mass} = n \times \text{gfm} = 0.2 \times 44 = 8.8\text{g}$ ☒ D $1\text{mol NH}_3 = (1 \times 14) + (3 \times 1) = 14 + 3 = 17\text{g} \therefore \text{mass} = n \times \text{gfm} = 0.2 \times 17 = 3.4\text{g}$								
10	B	59	☒ A carbon dioxide is a non-metal oxide and forms an acid when added to water ☒ B copper (II) oxide is insoluble in water and does not change the pH of water ☒ C sodium oxide is a metal oxide and forms an alkali when added to water ☒ D sulphur dioxide is a non-metal oxide and forms an acid when added to water								
11	B	46	☒ A hydrochloric acid + sodium carbonate $\longrightarrow$ sodium chloride + water + carbon dioxide ☒ B sodium chloride does not react with acids as it is a salt not a base ☒ C hydrochloric acid + sodium hydroxide $\longrightarrow$ sodium chloride + water ☒ D hydrochloric acid + sodium oxide $\longrightarrow$ sodium chloride + water								
12	C	88	☒ A longest chain has five carbons $\therefore$ name ends in pentane ☒ B longest chain has five carbons $\therefore$ name ends in pentane ☒ C 5 carbons in main chain (pentane), two methyl groups on C_2 and C_3 ☒ D numbering of carbons from right to left to give side groups lower numbering								

13	A	81	☒ A correct structure is drawn. ☒ B Shortened structural formula would be: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{OH})\text{C}(\text{CH}_3)_3$ ☒ C Shortened structural formula would be: $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}(\text{OH})\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$ ☒ D Shortened structural formula would be: $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$																									
14	D	77	Molecular formula of structures: C_4H_6 C_5H_8 C_6H_{10} <table><tr><th>Answer</th><th>General Formula</th><th>Formula if n=4</th><th>Formula if n=5</th><th>Formula if n=6</th></tr><tr><td>A</td><td>$\text{C}_n\text{H}_{2n-4}$</td><td>$\text{C}_4\text{H}_4$</td><td>$\text{C}_5\text{H}_6$</td><td>$\text{C}_6\text{H}_8$</td></tr><tr><td>B</td><td>$\text{C}_n\text{H}_{2n+2}$</td><td>$\text{C}_4\text{H}_{10}$</td><td>$\text{C}_5\text{H}_{12}$</td><td>$\text{C}_6\text{H}_{14}$</td></tr><tr><td>C</td><td>C_nH_{2n}</td><td>C_4H_8</td><td>C_5H_{10}</td><td>C_6H_{12}</td></tr><tr><td>D</td><td>$\text{C}_n\text{H}_{2n-2}$</td><td>$\text{C}_4\text{H}_6$</td><td>$\text{C}_5\text{H}_8$</td><td>$\text{C}_6\text{H}_{10}$</td></tr></table>	Answer	General Formula	Formula if n=4	Formula if n=5	Formula if n=6	A	$\text{C}_n\text{H}_{2n-4}$	C_4H_4	C_5H_6	C_6H_8	B	$\text{C}_n\text{H}_{2n+2}$	C_4H_{10}	C_5H_{12}	C_6H_{14}	C	C_nH_{2n}	C_4H_8	C_5H_{10}	C_6H_{12}	D	$\text{C}_n\text{H}_{2n-2}$	C_4H_6	C_5H_8	C_6H_{10}
Answer	General Formula	Formula if n=4	Formula if n=5	Formula if n=6																								
A	$\text{C}_n\text{H}_{2n-4}$	C_4H_4	C_5H_6	C_6H_8																								
B	$\text{C}_n\text{H}_{2n+2}$	C_4H_{10}	C_5H_{12}	C_6H_{14}																								
C	C_nH_{2n}	C_4H_8	C_5H_{10}	C_6H_{12}																								
D	$\text{C}_n\text{H}_{2n-2}$	C_4H_6	C_5H_8	C_6H_{10}																								
15	C	67	☒ A negative and positive ions are only found in ionic bonding ☒ B a shared pair of electrons and two nuclei are only found in covalent bonding ☒ C metallic bonding has positive ions (nucleus & inner shells) and delocalised electrons ☒ D metallic bonding has delocalised electrons not delocalised protons																									
16	D	76	☒ A zinc & tin give lower voltage as they are closer together on electrochemical series ☒ B electrons would flow from Y (zinc) to X (tin) ☒ C electrons would flow from Y (magnesium) to X (copper) ☒ D magnesium and copper would give the biggest voltage as they are further apart on electrochemical series and electrons flow from X (magnesium) to Y (copper) as magnesium is higher up ECS																									
17	A	79	propene $\begin{array}{ccccccc} \text{H} & \text{H} & & \text{H} & \text{H} & & \text{H} & \text{H} \\ & & & & & & & \\ \text{C} & = & \text{C} & + & \text{C} & = & \text{C} & + & \text{C} & = & \text{C} \\ & & & & & & & \\ \text{CH}_3 & \text{H} & & \text{CH}_3 & \text{H} & & \text{CH}_3 & \text{H} \end{array}$ $\downarrow$ poly(propene) $\begin{array}{ccccccc} \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} \\ & & & & & \\ -\text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C}- \\ & & & & & \\ \text{CH}_3 & \text{H} & \text{CH}_3 & \text{H} & \text{CH}_3 & \text{H} \end{array}$																									
17	A	79	<table><tr><td>$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{C} = \text{C} \\ \quad \\ \text{CH}_3 \quad \text{H} \end{array}$</td><td>$\begin{array}{ccccccc} \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} \\ & & & & & \\ -\text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C}- \\ & & & & & \\ \text{CH}_3 & \text{H} & \text{CH}_3 & \text{H} & \text{CH}_3 & \text{H} \end{array}$</td><td>$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ -\text{C} - \text{C}- \\ \quad \\ \text{CH}_3 \quad \text{H} \end{array}$</td></tr><tr><td>Monomer</td><td>Polymer</td><td>Repeating Unit</td></tr></table>	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{C} = \text{C} \\ \quad \\ \text{CH}_3 \quad \text{H} \end{array}$	$\begin{array}{ccccccc} \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} \\ & & & & & \\ -\text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C}- \\ & & & & & \\ \text{CH}_3 & \text{H} & \text{CH}_3 & \text{H} & \text{CH}_3 & \text{H} \end{array}$	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ -\text{C} - \text{C}- \\ \quad \\ \text{CH}_3 \quad \text{H} \end{array}$	Monomer	Polymer	Repeating Unit																			
$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{C} = \text{C} \\ \quad \\ \text{CH}_3 \quad \text{H} \end{array}$	$\begin{array}{ccccccc} \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} \\ & & & & & \\ -\text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C}- \\ & & & & & \\ \text{CH}_3 & \text{H} & \text{CH}_3 & \text{H} & \text{CH}_3 & \text{H} \end{array}$	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ -\text{C} - \text{C}- \\ \quad \\ \text{CH}_3 \quad \text{H} \end{array}$																										
Monomer	Polymer	Repeating Unit																										
18	A	38	Spectator ions appear chemically unchanged on both sides of a chemical equation: $2\text{Na}^+_{(\text{aq})} + \text{SO}_4^{2-}_{(\text{aq})} + \text{Ba}^{2+}_{(\text{aq})} + 2\text{Cl}^-_{(\text{aq})} \rightarrow \text{BaSO}_{4(\text{s})} + 2\text{Na}^+_{(\text{aq})} + \text{SO}_4^{2-}_{(\text{aq})}$ $\text{SO}_4^{2-} \text{ appears on both sides of equation}$																									
19	D	33	☒ A swap names: ammonium nitrate and potassium chloride are both very soluble ☒ B swap names: zinc sulphate and magnesium nitrate are both very soluble ☒ C swap names: calcium chloride and nickel chloride are both very soluble ☒ D swap names: sodium nitrate is very soluble but silver iodide is insoluble																									
20	D	61	Potassium chloride is colourless $\therefore$ potassium ions and chloride ions are colourless. Potassium chromate is yellow. As potassium ions are colourless, yellow colour must be due to the chromate ion.																									

2015 National 5 Chemistry Marking Scheme

Long Qu	Answer	Reasoning																									
1a	$0.8\text{ cm}^3\text{ s}^{-1}$	$\text{Rate} = \frac{\Delta\text{quantity}}{\Delta\text{time}} = \frac{120 - 96}{90 - 60} = \frac{24}{30} = 0.8\text{ cm}^3\text{ s}^{-1}$																									
1b	Graph showing:	Both axes labelled with units (1 mark) Both scales (1 mark) Graph drawn accurately (1mark) - points must be plotted correctly and line drawn, either by joining the dots or by a smooth curve or curve of best fit - The line must be drawn from the origin.																									
2a	Neptunium	${}_{95}^{241}\text{Am} \rightarrow {}_2^4\text{He} + {}_{93}^{237}\text{Np}$																									
2b	alpha	<table><tr><th>Radiation</th><th>Stopped by</th><th>Charge</th><th>Atomic Number</th><th>Mass Number</th></tr><tr><td>Alpha</td><td>paper</td><td>Positive</td><td>2</td><td>4</td></tr><tr><td>Beta</td><td>aluminium</td><td>Negative</td><td>-1</td><td>0</td></tr><tr><td>Gamma</td><td>lead</td><td>No charge</td><td colspan="2">Gamma radiation is a wave not a particle</td></tr></table>	Radiation	Stopped by	Charge	Atomic Number	Mass Number	Alpha	paper	Positive	2	4	Beta	aluminium	Negative	-1	0	Gamma	lead	No charge	Gamma radiation is a wave not a particle						
Radiation	Stopped by	Charge	Atomic Number	Mass Number																							
Alpha	paper	Positive	2	4																							
Beta	aluminium	Negative	-1	0																							
Gamma	lead	No charge	Gamma radiation is a wave not a particle																								
2c(i)	1g	<table><tr><th>Time (hr)</th><th>Mass (g)</th></tr><tr><td>0</td><td>8</td></tr><tr><td>16</td><td>4</td></tr><tr><td>32</td><td>2</td></tr><tr><td>48</td><td>1</td></tr></table>	Time (hr)	Mass (g)	0	8	16	4	32	2	48	1															
Time (hr)	Mass (g)																										
0	8																										
16	4																										
32	2																										
48	1																										
2c(ii)	Longer half-life	Americium-242 has too short a half-life to be effective in a smoke-detector as the amount of Americium would half every 16 hours. It would not be operational within days of manufacture. Americium-241 has a half-life of 432years and will be working for many lifetimes but must be disposed of carefully.																									
3a(i)	hydroxyl	Hydroxyl groups are the functional group found in all alcohols and have the structure -OH or -O-H																									
3a(ii)	Esters or fats or oils	Fats and oils are triglycerides with three ester bonds between the original glycerol molecule and three fatty acids																									
3b(i)	Butanoic acid	<table><tr><td>$\begin{array}{c} \text{O} \\ \parallel \\ \text{H}-\text{C} \\ \\ \text{OH} \end{array}$ methanoic acid </td><td>$\begin{array}{c} \text{H} & \text{O} \\ & \parallel \\ \text{H}-\text{C}-\text{C} \\ & \\ \text{H} & \text{OH} \end{array}$ ethanoic acid </td></tr><tr><td colspan="2"></td><td><table><tr><td>$\begin{array}{c} \text{H} & \text{H} & \text{O} \\ & & \parallel \\ \text{H}-\text{C}-\text{C}-\text{C} \\ & & \\ \text{H} & \text{H} & \text{OH} \end{array}$ propanoic acid </td><td>$\begin{array}{c} \text{H} & \text{H} & \text{H} & \text{O} \\ & & & \parallel \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C} \\ & & & \\ \text{H} & \text{H} & \text{H} & \text{OH} \end{array}$ butanoic acid </td></tr></table></td></tr><tr><td>3b(ii)</td><td>Bromine decolourises</td><td>Bromine solution is an yellow/orange colour contains molecules of Br₂. The solution will decolourise as the bromine adds across the double bond. Only C=C double bonds will decolourise bromine solution.</td></tr><tr><td>4a</td><td>Diagram showing:</td><td></td></tr><tr><td>4b</td><td>hydrogen hydroxide</td><td><table><tr><td>Acids</td><td colspan="2">Hydrogen ion concentration greater than hydroxide ion concentration</td></tr><tr><td>Neutral</td><td>Hydrogen ion concentration equal to</td><td>hydroxide ion concentration</td></tr><tr><td>Alkali</td><td>Hydrogen ion concentration less than</td><td>hydroxide ion concentration</td></tr></table></td></tr></table>	$\begin{array}{c} \text{O} \\ \parallel \\ \text{H}-\text{C} \\ \\ \text{OH} \end{array}$ methanoic acid	$\begin{array}{c} \text{H} & \text{O} \\ & \parallel \\ \text{H}-\text{C}-\text{C} \\ & \\ \text{H} & \text{OH} \end{array}$ ethanoic acid			<table><tr><td>$\begin{array}{c} \text{H} & \text{H} & \text{O} \\ & & \parallel \\ \text{H}-\text{C}-\text{C}-\text{C} \\ & & \\ \text{H} & \text{H} & \text{OH} \end{array}$ propanoic acid </td><td>$\begin{array}{c} \text{H} & \text{H} & \text{H} & \text{O} \\ & & & \parallel \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C} \\ & & & \\ \text{H} & \text{H} & \text{H} & \text{OH} \end{array}$ butanoic acid </td></tr></table>	$\begin{array}{c} \text{H} & \text{H} & \text{O} \\ & & \parallel \\ \text{H}-\text{C}-\text{C}-\text{C} \\ & & \\ \text{H} & \text{H} & \text{OH} \end{array}$ propanoic acid	$\begin{array}{c} \text{H} & \text{H} & \text{H} & \text{O} \\ & & & \parallel \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C} \\ & & & \\ \text{H} & \text{H} & \text{H} & \text{OH} \end{array}$ butanoic acid	3b(ii)	Bromine decolourises	Bromine solution is an yellow/orange colour contains molecules of Br ₂ . The solution will decolourise as the bromine adds across the double bond. Only C=C double bonds will decolourise bromine solution.	4a	Diagram showing:		4b	hydrogen hydroxide	<table><tr><td>Acids</td><td colspan="2">Hydrogen ion concentration greater than hydroxide ion concentration</td></tr><tr><td>Neutral</td><td>Hydrogen ion concentration equal to</td><td>hydroxide ion concentration</td></tr><tr><td>Alkali</td><td>Hydrogen ion concentration less than</td><td>hydroxide ion concentration</td></tr></table>	Acids	Hydrogen ion concentration greater than hydroxide ion concentration		Neutral	Hydrogen ion concentration equal to	hydroxide ion concentration	Alkali	Hydrogen ion concentration less than	hydroxide ion concentration
$\begin{array}{c} \text{O} \\ \parallel \\ \text{H}-\text{C} \\ \\ \text{OH} \end{array}$ methanoic acid	$\begin{array}{c} \text{H} & \text{O} \\ & \parallel \\ \text{H}-\text{C}-\text{C} \\ & \\ \text{H} & \text{OH} \end{array}$ ethanoic acid																										
		<table><tr><td>$\begin{array}{c} \text{H} & \text{H} & \text{O} \\ & & \parallel \\ \text{H}-\text{C}-\text{C}-\text{C} \\ & & \\ \text{H} & \text{H} & \text{OH} \end{array}$ propanoic acid </td><td>$\begin{array}{c} \text{H} & \text{H} & \text{H} & \text{O} \\ & & & \parallel \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C} \\ & & & \\ \text{H} & \text{H} & \text{H} & \text{OH} \end{array}$ butanoic acid </td></tr></table>	$\begin{array}{c} \text{H} & \text{H} & \text{O} \\ & & \parallel \\ \text{H}-\text{C}-\text{C}-\text{C} \\ & & \\ \text{H} & \text{H} & \text{OH} \end{array}$ propanoic acid	$\begin{array}{c} \text{H} & \text{H} & \text{H} & \text{O} \\ & & & \parallel \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C} \\ & & & \\ \text{H} & \text{H} & \text{H} & \text{OH} \end{array}$ butanoic acid																							
$\begin{array}{c} \text{H} & \text{H} & \text{O} \\ & & \parallel \\ \text{H}-\text{C}-\text{C}-\text{C} \\ & & \\ \text{H} & \text{H} & \text{OH} \end{array}$ propanoic acid	$\begin{array}{c} \text{H} & \text{H} & \text{H} & \text{O} \\ & & & \parallel \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C} \\ & & & \\ \text{H} & \text{H} & \text{H} & \text{OH} \end{array}$ butanoic acid																										
3b(ii)	Bromine decolourises	Bromine solution is an yellow/orange colour contains molecules of Br ₂ . The solution will decolourise as the bromine adds across the double bond. Only C=C double bonds will decolourise bromine solution.																									
4a	Diagram showing:																										
4b	hydrogen hydroxide	<table><tr><td>Acids</td><td colspan="2">Hydrogen ion concentration greater than hydroxide ion concentration</td></tr><tr><td>Neutral</td><td>Hydrogen ion concentration equal to</td><td>hydroxide ion concentration</td></tr><tr><td>Alkali</td><td>Hydrogen ion concentration less than</td><td>hydroxide ion concentration</td></tr></table>	Acids	Hydrogen ion concentration greater than hydroxide ion concentration		Neutral	Hydrogen ion concentration equal to	hydroxide ion concentration	Alkali	Hydrogen ion concentration less than	hydroxide ion concentration																
Acids	Hydrogen ion concentration greater than hydroxide ion concentration																										
Neutral	Hydrogen ion concentration equal to	hydroxide ion concentration																									
Alkali	Hydrogen ion concentration less than	hydroxide ion concentration																									

4c	Answer to include:	1 st Mark: Calcium oxide as a base or forms an alkali when dissolved in water (must mention when dissolved in water) 2 nd Mark: Calcium oxide in water <u>neutralises</u> sulphur dioxide (must mention the word neutralise)							
5a	Iron	Iron is the catalyst in the Haber Process where nitrogen and hydrogen react to become ammonia NH ₃ .							
5b	52-56	Pressure	100	200	300	400	500	600	700
		% Yield	10	18	26	32	40	-	-
		Difference	8	8	8	8	(8)	(8)	
		Estimate	-	-	-	-	-	48	56
5c	Higher the temp the lower the yield	The Haber Process is a reversible reaction where equilibrium is eventually achieved where the rate of the formation of products equals the rate of the breakdown of products.							
5d	Low Temperature High Pressure	In top table, the higher the pressure the higher the yield. In bottom table, the lower the temperature the higher the yield							
6a	3	99% of carbon dioxide is captured ∴ 1% is released $1\% \text{ of } 300\text{tonnes} = \frac{1}{100} \times 300 = 3\text{tonnes}$							
6b	Fe ₂ O ₃	Write down Valency below each element's symbol Fe O 3 2		Put in Cross-over Arrows Fe O ↙ ↘ 3 2		Follow arrows to get formula Fe ₂ O ₃			
6c	exothermic	Exothermic	Chemical reaction which gives out energy						
		Endothermic	Chemical reaction which takes in energy from the surroundings						
7a	Boil off the water (using a Bunsen burner)	The solution of copper sulphate is boiled in an evaporating basin until almost dry. The Bunsen burner is turned off just before dryness and the remaining heat evaporated the remaining water.							
7b	0.2	$1\text{mol CuSO}_4 = (1 \times 63.5) + (1 \times 32) + (4 \times 16) = 63.5 + 32 + 64 = 159.5\text{g}$ $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{3.19}{159.5} = 0.02\text{mol}$ $\text{concentration} = \frac{\text{no. of mol}}{\text{volume}} = \frac{0.02}{0.1} = 0.2 \text{ mol l}^{-1}$							
8a	One answer from:	Method A			Method B				
		incomplete combustion/ Less oxygen			complete combustion/more oxygen				
		(more) heat loss to surroundings			less/no heat loss to surroundings				
		no draught shield/no insulation			better insulation				
		glass is a poor conductor			metal/platinum is a better conductor				
		flame too far away from beaker							
8b	14.2 kJ	E _h	=	c	×	m	×	ΔT	
		Energy	=	specific heat capacity	×	mass	×	change in temperature	
		E _h	=	4.18	×	0.1	×	34	
		E _h	=	14.212kJ					
9a	ore	Ores are compounds of metals from which the metal can be extracted.							
9b(i)	$4\text{Al}^{3+} + 6\text{O}^{2-}$ $\downarrow$ $4\text{Al} + 3\text{O}_2$	$4\text{Al}^{3+} + 6\text{O}^{2-} \longrightarrow 4\text{Al} + 3\text{O}_2$							

9b(ii)	Ions free/able to move	Solid ionic compounds are unable to conduct as the ions are locked tightly in an ionic lattice. On melting or dissolving in water, ions become free to move and will conduct.																		
9c	Magnesium	<table><tr><td>Agent</td><td>Oxidation/Reduction</td><td>Electrons</td></tr><tr><td>Oxidising Agent</td><td>Agent is reduced itself</td><td>Agent gains electrons</td></tr><tr><td>Reducing Agent</td><td>Agent is oxidised itself</td><td>Agent loses electrons</td></tr></table> $\text{Mg} \longrightarrow \text{Mg}^{2+} + 2\text{e}^{-}$ Magnesium loses electrons, is oxidised itself and acts as a reducing agent	Agent	Oxidation/Reduction	Electrons	Oxidising Agent	Agent is reduced itself	Agent gains electrons	Reducing Agent	Agent is oxidised itself	Agent loses electrons									
Agent	Oxidation/Reduction	Electrons																		
Oxidising Agent	Agent is reduced itself	Agent gains electrons																		
Reducing Agent	Agent is oxidised itself	Agent loses electrons																		
10	Open Question Answer to include:	<table><tr><td>3 mark answer Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>2 mark answer Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>1 mark answer Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table>	3 mark answer Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	2 mark answer Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	1 mark answer Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.															
3 mark answer Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	2 mark answer Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	1 mark answer Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.																		
11a	2,8,6	$\underset{2,8,7}{\text{Cl(g)}} \longrightarrow \underset{2,8,6}{\text{Cl}^{+}(\text{g})} + \text{e}^{-}$																		
11b	$\text{Mg(g)} \longrightarrow \text{Mg}^{+}(\text{g}) + \text{e}^{-}$	The symbol for Cl in the example must be substituted with Mg for Magnesium. This must always be in the gas state,																		
11c	decreases	Alkali Metals in group 1: lithium, sodium and potassium: - as you go down group 1 ionisation energy decreases. <table><tr><td>Alkali Metal</td><td>Li</td><td>Na</td><td>K</td></tr><tr><td>First Ionisation Energy (kJ mol⁻¹)</td><td>526</td><td>502</td><td>425</td></tr></table> Halogens in group 7: fluorine, chlorine and bromine: - As you go down group 7 ionisation energy decreases. <table><tr><td>Halogen</td><td>F</td><td>Cl</td><td>Br</td></tr><tr><td>First Ionisation Energy (kJ mol⁻¹)</td><td>1690</td><td>1260</td><td>1150</td></tr></table>			Alkali Metal	Li	Na	K	First Ionisation Energy (kJ mol ⁻¹)	526	502	425	Halogen	F	Cl	Br	First Ionisation Energy (kJ mol ⁻¹)	1690	1260	1150
Alkali Metal	Li	Na	K																	
First Ionisation Energy (kJ mol ⁻¹)	526	502	425																	
Halogen	F	Cl	Br																	
First Ionisation Energy (kJ mol ⁻¹)	1690	1260	1150																	
12a	but-2-ene	but - 2 - ene 4 carbons C=C on 2nd carbon C=C double bond																		
12b	Same formula but different structure	Isomer: same chemical formula but different structural formula. Both structures have the formula C ₄ H ₈ but belong to different homologous series. Structure A is an alkene with a C=C double bond and Structure B is a cycloalkane with a ring of carbons.																		
12c	Any structure of 3-methylpent-2-ene or 2-ethylbut-1-ene	H H-C H H C=C H H C H H C H H C H 3-methylpent-2-ene or H H-C H H C H H C H H C H H C H 2-ethylbut-1-ene																		
13a	Carboxyl group	The carboxyl group -COOH is the functional group found in carboxylic acids. Succinic acid contains two carboxyl groups. O -C-OH carboxyl group																		

13b(i)	Condensation	Condensation reactions take place between carboxyl groups and hydroxyl groups with water removed as the groups join together. $\begin{array}{c} \text{O} \\ \\ -\text{C}-\text{OH} \\ \text{carboxyl group} \end{array} + \begin{array}{c} \text{H}-\text{O}- \\ \text{hydroxyl group} \end{array} \xrightarrow[\text{water removed at join}]{\text{condensation}} \begin{array}{c} \text{O} \\ \\ -\text{C}-\text{O}- \\ \text{ester link} \end{array}$						
13b(ii)	Diagram showing:	$\begin{array}{ccccccc} & \text{H} & \text{H} & & \text{O} & \text{H} & \text{H} & & \text{O} \\ & & & & & & & & \\ -\text{O} & -\text{C} & -\text{C} & -\text{O} & -\text{C} & -\text{C} & -\text{C} & -\text{C}- \\ & & & & & & & & \\ & \text{H} & \text{H} & & & \text{H} & \text{H} & & \end{array}$						
14a(i)	Carbon monoxide	$\text{TiO}_2 + 2\text{Cl}_2 + 2\text{C} \rightarrow \text{TiCl}_4 + 2\text{CO}$						
14a(ii)	Covalent	As titanium (IV) chloride is a compound of a metal and non-metal, your first assumption would be that it contains ionic bonding. However, ionic compounds are always solids at room temperature with high melting points and conduct electricity when in the liquid/molten state. Titanium (IV) chloride contains covalent bonding as it is a liquid at room temperature and does not conduct electricity in the liquid state. Metallic bonding can be ruled out as it does not conduct electricity in the liquid state.						
14b	Distillation	Distillation separates titanium (IV) chloride from its impurities as distillation separates chemicals with different boiling points. The titanium (IV) chloride would evaporate and then condensed in the distillation apparatus leaving the impurities behind.						
14c	recycle/reuse sodium or chlorine	Electrolysing molten sodium chloride would produce sodium metal at the negative electrode and chlorine gas at the positive electrode. Chlorine gas is needed for the reaction where TiO_2 is converted into TiCl_4 . Sodium metal is required to displacement reaction to extract Ti from TiCl_4 .						
15a	16	Ignore Titration 1 as it is the rough titre and is not designed to be accurate $\text{Average titre} = \frac{15.9 + 16.1}{2} = \frac{32.0}{2} = 16.0 \text{ cm}^3$						
15b	0.0032	no. of moles = volume x concentration = 0.016litres x 0.005mol l ⁻¹ = 0.00008 mol $\begin{array}{ccccccc} \text{C}_6\text{H}_8\text{O}_6 & + & \text{I}_2 & \rightarrow & \text{C}_6\text{H}_6\text{O}_6 & + & 2\text{HI} \\ 1\text{mol} & & 1\text{mol} & & & & \\ 0.00008\text{mol} & & 0.00008\text{mol} & & & & \end{array}$ $\text{concentration} = \frac{\text{no. of mol}}{\text{volume}} = \frac{0.00008\text{mol}}{0.025\text{litres}} = 0.0032 \text{ mol l}^{-1}$						
16	Open Question answer containing:	<table border="1"> <thead> <tr> <th>3 mark answer</th><th>2 mark answer</th><th>1 mark answer</th></tr> </thead> <tbody> <tr> <td>Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr> </tbody> </table>	3 mark answer	2 mark answer	1 mark answer	Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.
3 mark answer	2 mark answer	1 mark answer						
Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.						

2015 National 5 Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D
1	A	92	3	2	2
2	B	4	88	4	4
3	D	2	15	16	67
4	C	3	26	55	15
5	D	8	8	6	78
6	C	21	15	51	11
7	C	27	13	46	14
8	B	5	90	4	1
9	A	69	6	12	11
10	B	21	59	12	7
11	B	18	46	18	17
12	C	4	1	88	7
13	A	81	6	6	6
14	D	1	9	13	77
15	C	18	10	67	5
16	D	7	4	14	76
17	A	79	17	1	2
18	A	38	9	8	44
19	D	22	24	19	33
20	D	6	20	13	61